

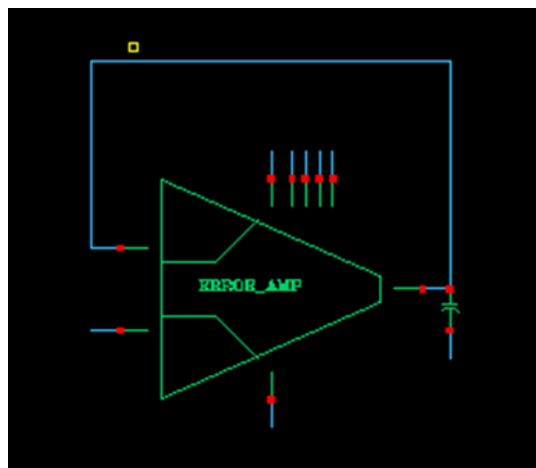
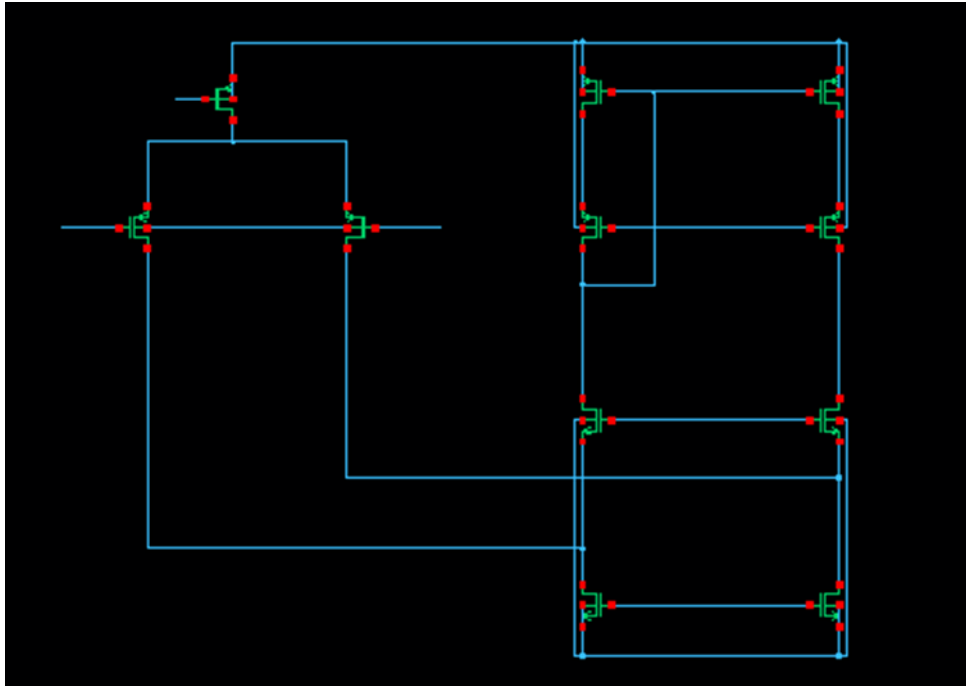
Họ tên: Lê Minh Hiền

Trường: Đại học Công Nghiệp Hà Nội

Thời gian: 01/08/2026

Thiết kế mạch khuếch đại thuật toán trên Cadence Virtuoso công nghệ CMOS 45nm.

Folded Cascode Amp 45nm



$$A_v = 54 \text{ dB} = 533 \Rightarrow g_{m1} \cdot (H_1 \parallel H_2)$$

$$H_1 = g_{m3} \cdot r_{o3} \cdot (r_{o3} \parallel r_{o2})$$

$$H_2 = g_{m4} \cdot r_{o4} \cdot r_{o5}$$

$$H_1 = 3H_2$$

$$\frac{2}{3} g_{m1} H_1 = 533$$

$$\Rightarrow g_{m1} \cdot g_{m3} \cdot r_{o3} \cdot (r_{o3} \parallel r_{o2}) = 800$$

$$\Rightarrow g_{m1} r_{o1} = 2r_{o2}$$

$$\Rightarrow \frac{g_{m1} \cdot g_{m3} \cdot r_{o3} \cdot r_{o1}}{2} = 800$$

$$\Rightarrow g_{m1} \cdot g_{m3} \cdot r_{o3} \cdot r_{o1} = 1600$$

$$\Rightarrow g_{m1} \cdot r_{o1} = 60$$

$$g_{m3} \cdot r_{o3} = 40$$

$$\frac{1}{3} g_{m2} \cdot H_2 = 533$$

$$g_{m2} \cdot H_2 = 1599$$

$$\Rightarrow g_{m2} \cdot g_{m4} \cdot r_{o4} \cdot r_{o5} = 1599$$

$$\Rightarrow g_{m2} \cdot r_{o5} = g_{m4} \cdot r_{o4} = 40$$

$$N_{MUGBW} = 40 \text{ M} = \frac{g_{m1}}{2\pi \cdot C_L} \Rightarrow g_{m1} = 40 \text{ M} \cdot 10^{-12} \cdot 2\pi = 62,8 \mu \text{ S}$$

$$I_{SS} = 5,46 \mu \text{ A} \Rightarrow I_D = 2,73 \mu \text{ A} \Rightarrow g_{m1}/I_D = 23; V_{DS} = 0,6$$

$$g_{m1} \cdot r_{o1} = 60 \Rightarrow L = 0,8 \mu \text{ m}; V_{GS} = 0,4; I_D/w = 729 \mu \text{ A}; w = 3,75 \mu \text{ m}$$

$$M2: g_{ds2} = 2g_{ds1} = 2,1 \mu \text{ S}; I_D = 2,73 \mu \text{ A}; \lambda_L = \frac{g_{ds2}}{I_D} = 769 \mu \text{ A}; V_{GS} = 0,2$$

$$\Rightarrow L = 1,4 \mu \text{ m}; V_{GS} = 0,397 \text{ V}; I_D/w = 24,7 \mu \text{ A}; w = 0,26 \mu \text{ m}, 3 \mu \text{ m}$$

$$g_{m1}/I_D = 1,8$$

$$M5: g_{m2} \cdot r_{o5} = 40 \Rightarrow g_{ds5} = 1,57 \mu \text{ S}; I_D = 2,73 \mu \text{ A}$$

$$g_{m2} = 62,8 \mu \text{ S} \Rightarrow \lambda_5 = 575 \mu \text{ A}; V_{DS} = 0,3$$

$$\Rightarrow L = 0,5 \mu \text{ m}; V_{GS} = 0,396 \text{ V}; g_{m1}/I_D = 24,7; I_D/w = 716 \mu \text{ A}$$

$$\Rightarrow w = 3,8 \mu \text{ m}$$

$$M4: g_{m1} \cdot r_{o4} = 40 \Rightarrow L = 0,6 \mu \text{ m}; V_{GS} = 0,396 \text{ V}; g_{m1}/I_D = 24,7; I_D/w = 716 \mu \text{ A}$$

$$I_D/w = 263,5 \mu \text{ A}; w = 1,94 \mu \text{ m}; V_{DS} = 0,2$$

$$M3: g_{m2} \cdot r_{o3} = 40; V_{DS} = 0,5 \text{ V} \Rightarrow L = 0,7 \mu \text{ m}; V_{GS} = 0,4; I_D/w = 263,5 \mu \text{ A}$$

$$g_{m1}/I_D = 19; I_D/w = 795 \mu \text{ A} \Rightarrow w = 4,4 \mu \text{ m}$$

Gmoverid:

Name	Type	Details	Value	Plot	Save	Spec
gmoverid_PM1_1	expr	abs(OP("/IO/PM1_1" "gmoverid"))	28.83	☒	☐	
gmoverid_PM1_2	expr	abs(OP("/IO/PM1_2" "gmoverid"))	28.83	☒	☐	
gmoverid_NM2_1	expr	OP("/IO/NM2_1" "gmoverid")	18.37	☒	☐	
gmoverid_NM2_2	expr	OP("/IO/NM2_2" "gmoverid")	18.37	☒	☐	
gmoverid_NM3_1	expr	OP("/IO/NM3_1" "gmoverid")	19.7	☒	☐	
gmoverid_NM3_2	expr	OP("/IO/NM3_2" "gmoverid")	19.7	☒	☐	
gmoverid_PM4_1	expr	abs(OP("/IO/PM4_1" "gmoverid"))	26.12	☒	☐	
gmoverid_PM4_2	expr	abs(OP("/IO/PM4_2" "gmoverid"))	26.12	☒	☐	
gmoverid_PM5_1	expr	abs(OP("/IO/PM5_1" "gmoverid"))	24.28	☒	☐	
gmoverid_PM5_2	expr	abs(OP("/IO/PM5_2" "gmoverid"))	24.28	☒	☐	
gmoverid_PM6_1	expr	abs(OP("/IO/PM6" "gmoverid"))	21.92	☒	☐	
gm_PM1_1	expr	abs(OP("/IO/PM1_1" "gm"))	73.4u	☒	☐	
gm_PM1_2	expr	abs(OP("/IO/PM1_2" "gm"))	73.4u	☒	☐	
gm_NM2_1	expr	OP("/IO/NM2_1" "gm")	100u	☒	☐	
gm_NM2_2	expr	OP("/IO/NM2_2" "gm")	100u	☒	☐	
gm_NM3_1	expr	OP("/IO/NM3_1" "gm")	57.14u	☒	☐	
gm_NM3_2	expr	OP("/IO/NM3_2" "gm")	57.14u	☒	☐	
gm_PM4_1	expr	abs(OP("/IO/PM4_1" "gm"))	75.77u	☒	☐	
gm_PM4_2	expr	abs(OP("/IO/PM4_2" "gm"))	75.77u	☒	☐	
gm_PM5_1	expr	abs(OP("/IO/PM5_1" "gm"))	70.44u	☒	☐	
gm_PM5_2	expr	abs(OP("/IO/PM5_2" "gm"))	70.44u	☒	☐	

gm_PM6	expr	abs(OP("/IO/PM6" "gm"))	111.6u	☒	☐	
rout_PM1_1	expr	abs(OP("/IO/PM1_1" "rout"))	806.9k	☒	☐	
rout_PM1_2	expr	abs(OP("/IO/PM1_2" "rout"))	806.9k	☒	☐	
rout_NM2_1	expr	OP("/IO/NM2_1" "rout")	220.9k	☒	☐	
rout_NM2_2	expr	OP("/IO/NM2_2" "rout")	220.9k	☒	☐	
rout_NM3_1	expr	OP("/IO/NM3_1" "rout")	938.8k	☒	☐	
rout_NM3_2	expr	OP("/IO/NM3_2" "rout")	938k	☒	☐	
rout_PM4_1	expr	abs(OP("/IO/PM4_1" "rout"))	594.5k	☒	☐	
rout_PM4_2	expr	abs(OP("/IO/PM4_2" "rout"))	595.3k	☒	☐	
rout_PM5_1	expr	abs(OP("/IO/PM5_1" "rout"))	384.1k	☒	☐	
rout_PM5_2	expr	abs(OP("/IO/PM5_2" "rout"))	384.2k	☒	☐	
rout_PM6	expr	abs(OP("/IO/PM6" "rout"))	393.3k	☒	☐	
ids_PM1_1	expr	abs(OP("/IO/PM1_1" "ids"))	2.546u	☒	☐	
ids_PM1_2	expr	abs(OP("/IO/PM1_2" "ids"))	2.546u	☒	☐	
ids_NM2_1	expr	OP("/IO/NM2_1" "ids")	5.447u	☒	☐	
ids_NM2_2	expr	OP("/IO/NM2_2" "ids")	5.447u	☒	☐	
ids_NM3_1	expr	OP("/IO/NM3_1" "ids")	2.901u	☒	☐	
ids_NM3_2	expr	OP("/IO/NM3_2" "ids")	2.901u	☒	☐	
ids_PM4_1	expr	abs(OP("/IO/PM4_1" "ids"))	2.901u	☒	☐	
ids_PM4_2	expr	abs(OP("/IO/PM4_2" "ids"))	2.901u	☒	☐	
ids_PM5_1	expr	abs(OP("/IO/PM5_1" "ids"))	2.901u	☒	☐	

ids_PM5_2	expr	abs(OP("/IO/PM5_2" "ids"))	2.901u	☒	☐	
ids_PM6	expr	abs(OP("/IO/PM6" "ids"))	5.093u	☒	☐	

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒
Gm_over_id	gmoverid_PM1_1				21.78	37.02
Gm_over_id	gmoverid_PM1_2				21.78	37.02
Gm_over_id	gmoverid_NM2_1				16.29	21.2
Gm_over_id	gmoverid_NM2_2				16.29	21.2
Gm_over_id	gmoverid_NM3_1				17.57	21.92
Gm_over_id	gmoverid_NM3_2				17.57	21.92

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	gmoverid_PM4_1				19.59	33
Gm_over_id	gmoverid_PM4_2				19.59	33
Gm_over_id	gmoverid_PM5_1				18.57	30.39
Gm_over_id	gmoverid_PM5_2				18.57	30.39
Gm_over_id	gmoverid_PM6_1				16.01	28.01
Gm_over_id	gm_PM1_1				52.36u	98.64u
Gm_over_id	gm_PM1_2				52.36u	98.64u

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	gm_NM2_1				84.73u	122.6u
Gm_over_id	gm_NM2_2				84.73u	122.6u
Gm_over_id	gm_NM3_1				48.32u	69.5u
Gm_over_id	gm_NM3_2				48.32u	69.5u
Gm_over_id	gm_PM4_1				54.25u	105.1u
Gm_over_id	gm_PM4_2				54.25u	105.1u

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	gm_PM5_1				50.97u	96.38u
Gm_over_id	gm_PM5_2				50.97u	96.38u
Gm_over_id	gm_PM6				76.72u	147.8u
Gm_over_id	rout_PM1_1				630.8k	1.072M
Gm_over_id	rout_PM1_2				630.8k	1.072M
Gm_over_id	rout_NM2_1				195.3k	231.8k
Gm_over_id	rout_NM2_2				195.3k	231.8k

Test	Output	Spec	Weight	Pass/Fail	Min	Max	PVT_0
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	rout_NM3_1				694.6k	1.244M	730.6k
Gm_over_id	rout_NM3_2				694k	1.243M	730k
Gm_over_id	rout_PM4_1				217.5k	682.2k	529.9k
Gm_over_id	rout_PM4_2				219.1k	682.6k	530.2k
Gm_over_id	rout_PM5_1				231.3k	667.8k	254.7k
Gm_over_id	rout_PM5_2				231.3k	667.8k	254.7k
Gm_over_id	rout_PM6				86.96k	602k	146.5k

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	ids_PM1_1				2.069u	2.789u
Gm_over_id	ids_PM1_2				2.069u	2.789u
Gm_over_id	ids_NM2_1				4.8u	5.933u
Gm_over_id	ids_NM2_2				4.8u	5.933u
Gm_over_id	ids_NM3_1				2.712u	3.23u
Gm_over_id	ids_NM3_2				2.712u	3.23u

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Gm_over_id	ids_NM2_2				4.0u	3.555u
Gm_over_id	ids_NM3_1				2.712u	3.23u
Gm_over_id	ids_NM3_2				2.712u	3.23u
Gm_over_id	ids_PM4_1				2.712u	3.23u
Gm_over_id	ids_PM4_2				2.712u	3.23u
Gm_over_id	ids_PM5_1				2.712u	3.23u
Gm_over_id	ids_PM5_2				2.712u	3.23u
Gm_over_id	ids_PM6				4.139u	5.578u

Condition:

Name	Type	Details	Value	Plot	Save	Spec
vgs-vth PM1_1	expr	(abs(OP"/I/O/PM1_1" "vgs")) - abs(OP"/I/O/PM1_1" "vth"))	-22.49m	☒	☐	> -100m
vgs-vth NM2_1	expr	(OP"/I/O/NM2_1" "vgs") - OP"/I/O/NM2_1" "vth"))	18.31m	☒	☐	> -100m
vgs-vth NM3_1	expr	(OP"/I/O/NM3_1" "vgs") - OP"/I/O/NM3_1" "vth"))	-9.566m	☒	☐	> -100m
vgs-vth PM4_1	expr	(abs(OP"/I/O/PM4_1" "vgs")) - abs(OP"/I/O/PM4_1" "vth"))	35.06m	☒	☐	> -100m
vgs-vth PM5_1	expr	(abs(OP"/I/O/PM5_1" "vgs")) - abs(OP"/I/O/PM5_1" "vth"))	44.9m	☒	☐	> -100m
vgs-vth PM6_1	expr	(abs(OP"/I/O/PM6" "vgs")) - abs(OP"/I/O/PM6" "vth"))	77.36m	☒	☐	> -100m
vds-vdsat PM1_1	expr	(abs(OP"/I/O/PM1_1" "vds")) - abs(OP"/I/O/PM1_1" "vdsat"))	717.5m	☒	☐	> 10m
vds-vdsat NM2_1	expr	(OP"/I/O/NM2_1" "vds") - OP"/I/O/NM2_1" "vdsat"))	96.14m	☒	☐	> 10m
vds-vdsat NM3_1	expr	(OP"/I/O/NM3_1" "vds") - OP"/I/O/NM3_1" "vdsat"))	541.4m	☒	☐	> 10m
vds-vdsat PM4_1	expr	(abs(OP"/I/O/PM4_1" "vds")) - abs(OP"/I/O/PM4_1" "vdsat"))	173.3m	☒	☐	> 10m
vds-vdsat PM5_1	expr	(abs(OP"/I/O/PM5_1" "vds")) - abs(OP"/I/O/PM5_1" "vdsat"))	97.91m	☒	☐	> 10m
vds-vdsat PM6_1	expr	(abs(OP"/I/O/PM6" "vds")) - abs(OP"/I/O/PM6" "vdsat"))	166.4m	☒	☐	> 10m

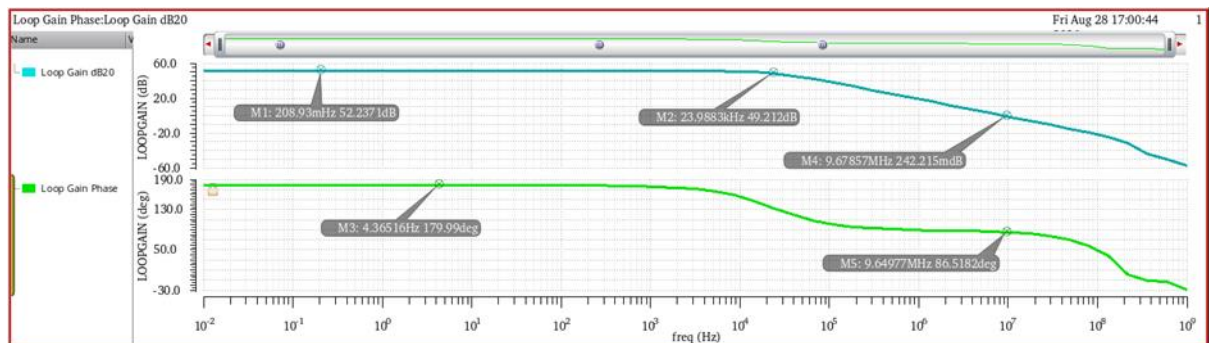
Parameter
VDD
gpdK045.scs
temperature

12 rows

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
condition	vgs-vth PM1_1	> -100m		pass	-34.84m	-4.743m
condition	vgs-vth NM2_1	> -100m		pass	9.07m	43m
condition	vgs-vth NM3_1	> -100m		pass	-13.94m	10.65m
condition	vgs-vth PM4_1	> -100m		pass	14.78m	66.99m
condition	vgs-vth PM5_1	> -100m		pass	23.99m	78.12m
condition	vgs-vth PM6_1	> -100m		pass	48.69m	123.7m
condition	vds-vdsat PM1_1	> 10m		pass	598.6m	818.5m
condition	vds-vdsat NM2_1	> 10m		pass	80.01m	106m
condition	vds-vdsat NM3_1	> 10m		pass	370.3m	708.2m

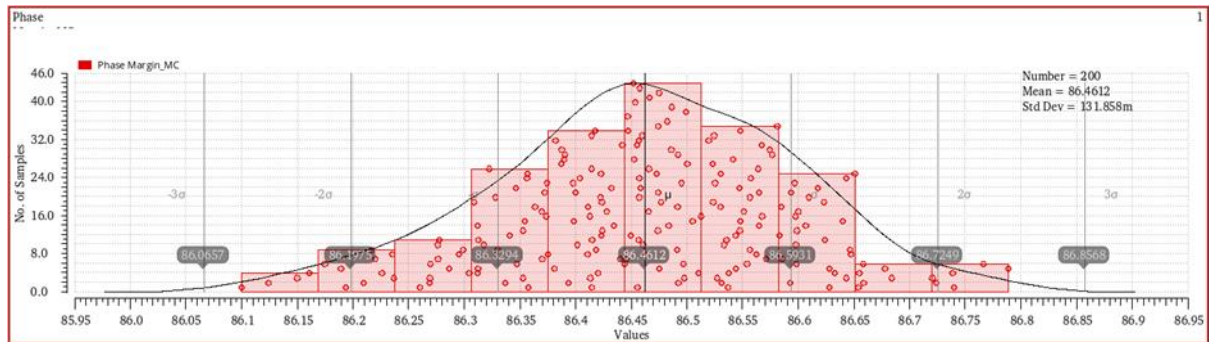
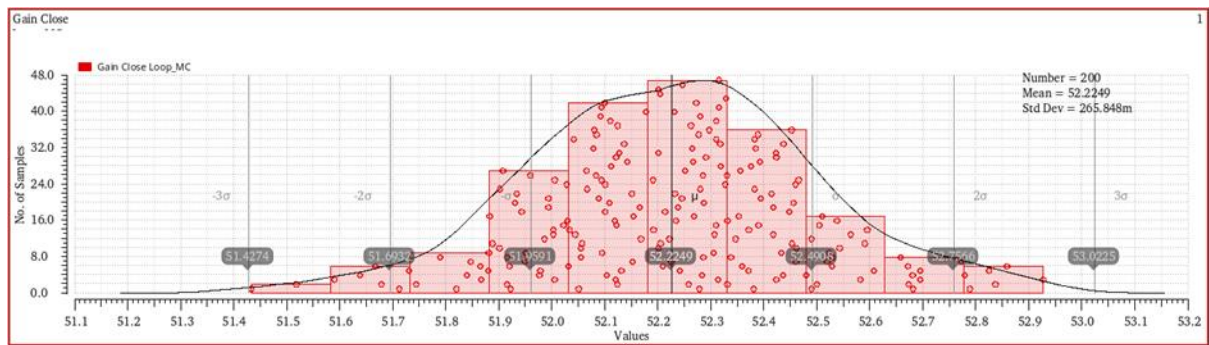
condition	vgs-vth PM4_1	> -100m		pass	14.78m	66.99m
condition	vgs-vth PM5_1	> -100m		pass	23.99m	78.12m
condition	vgs-vth PM6_1	> -100m		pass	48.69m	123.7m
condition	vds-vdsat PM1_1	> 10m		pass	598.6m	818.5m
condition	vds-vdsat NM2_1	> 10m		pass	80.01m	106m
condition	vds-vdsat NM3_1	> 10m		pass	370.3m	708.2m
condition	vds-vdsat PM4_1	> 10m		pass	43.05m	274.6m
condition	vds-vdsat PM5_1	> 10m		pass	74.48m	136.7m
condition	vds-vdsat PM6_1	> 10m		pass	27.07m	310.7m

Close Loop:

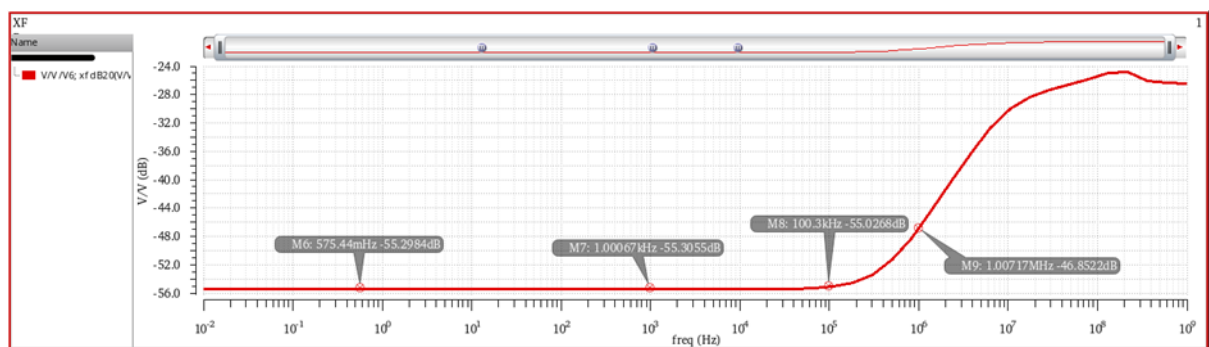


Gain Loop	expr	value(db(maggetData("loopGain"?result...	52.19	☒	☐	
Phase Margin	expr	getData("phaseMargin"?result"stb_margi...	86.46	☒	☐	
Bandwidth	expr	cross(db(maggetData("loopGain"?result"...	23.99k	☒	☐	
Gain Bandwidth	expr	cross(db(maggetData("loopGain"?result"...	9.947M	☒	☐	

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒
close_loop	Gain Loop				49	53.92
close_loop	Phase Margin				85.87	86.98
close_loop	Bandwidth				22.59k	28.27k
close_loop	Gain Bandwidth				7.039M	13.67M



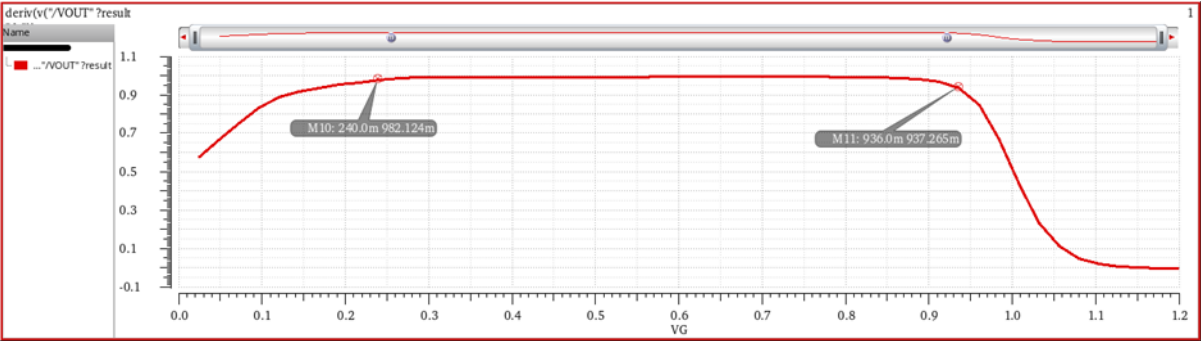
Psrr:



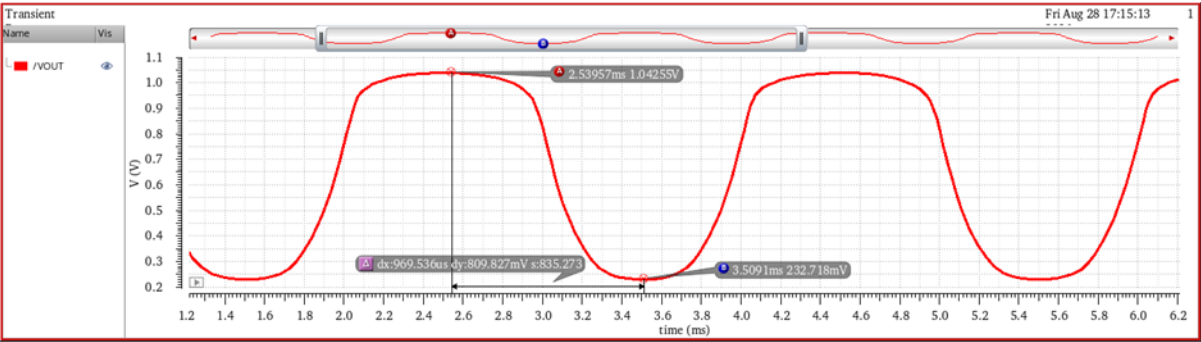
Psrr 0Hz	expr	value(db(getData("/V1" ?result "xf")) 0)	-55.24	☒	☐
Psrr 1kHz	expr	value(db(getData("/V1" ?result "xf")) 1000)	-55.24	☒	☐
Psrr 1MHz	expr	value(db(getData("/V1" ?result "xf")) 1000000)	-46.9	☒	☐
Psrr 10MHz	expr	value(db(getData("/V1" ?result "xf")) 10000000)	-30.23	☒	☐

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
psrr	Psrr 0Hz				-57.2	-50.83
psrr	Psrr 1kHz				-57.2	-50.83
psrr	Psrr 1MHz				-49.71	-42.59
psrr	Psrr 10MHz				-32.23	-27.6

Icmr:

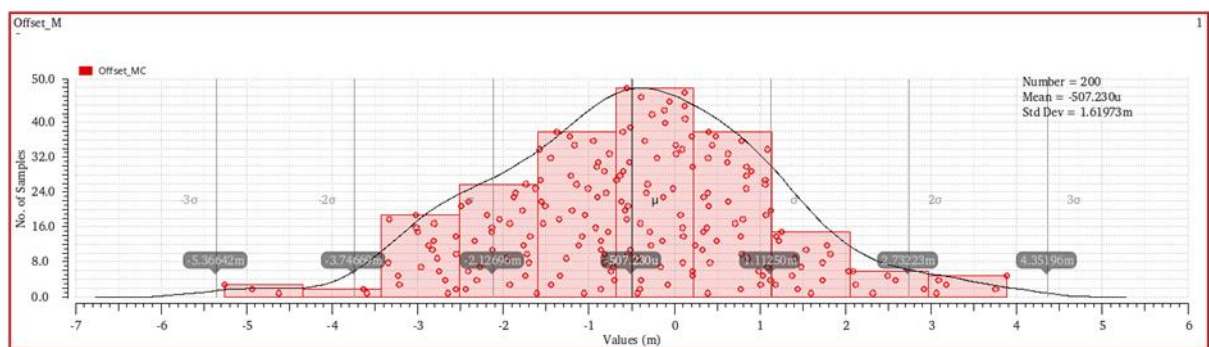
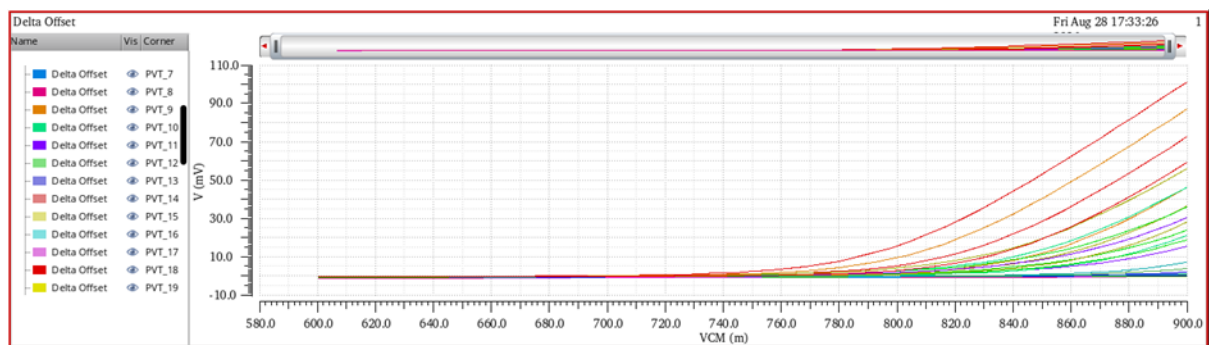
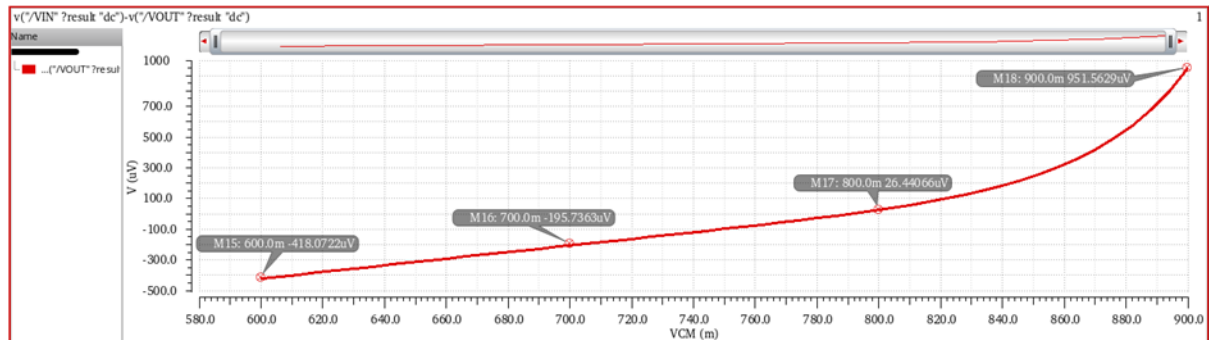
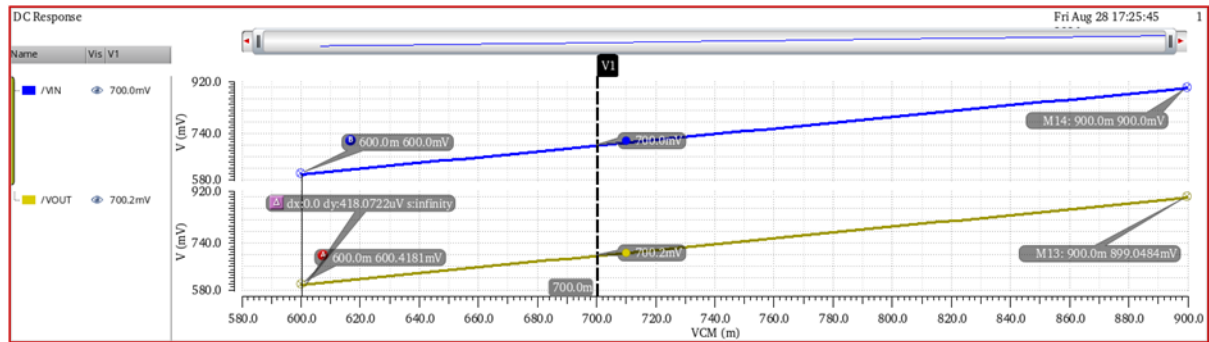


Output_swing:



Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
Test:folded:1	/VOUT					
Test:folded:1	Peak to Peak				571.9m	1.004

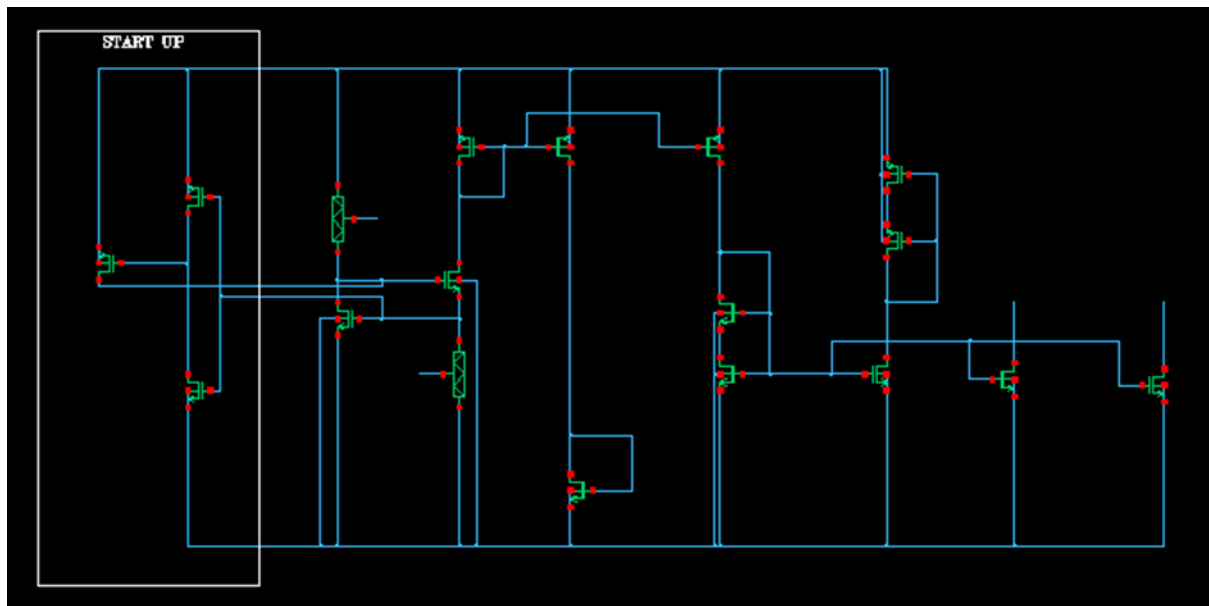
Offset:



Slew rate:

Slew rate expr slewRate(v("/VOUT" ?result "tran") 0.03663 ... 3.838M ☒ ☐

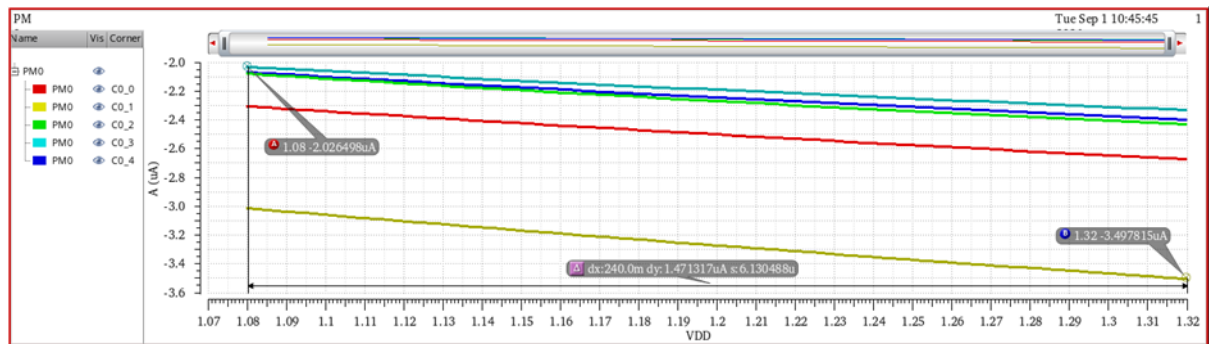
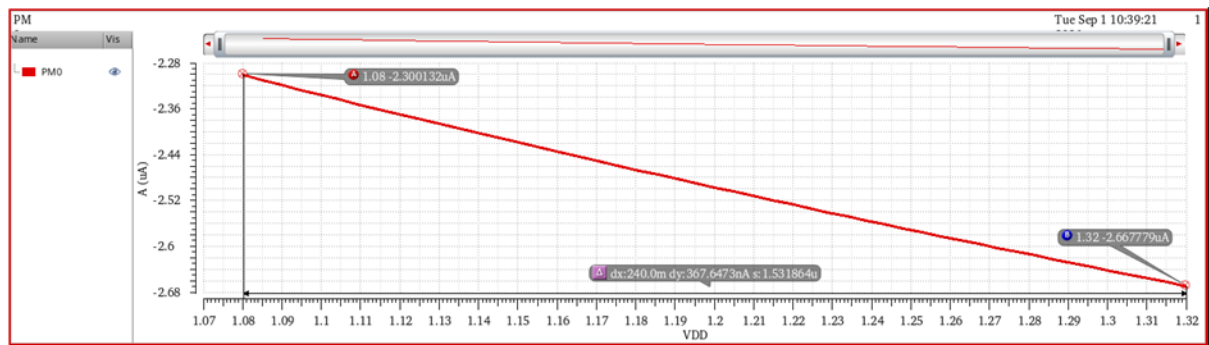
Bias 45nm



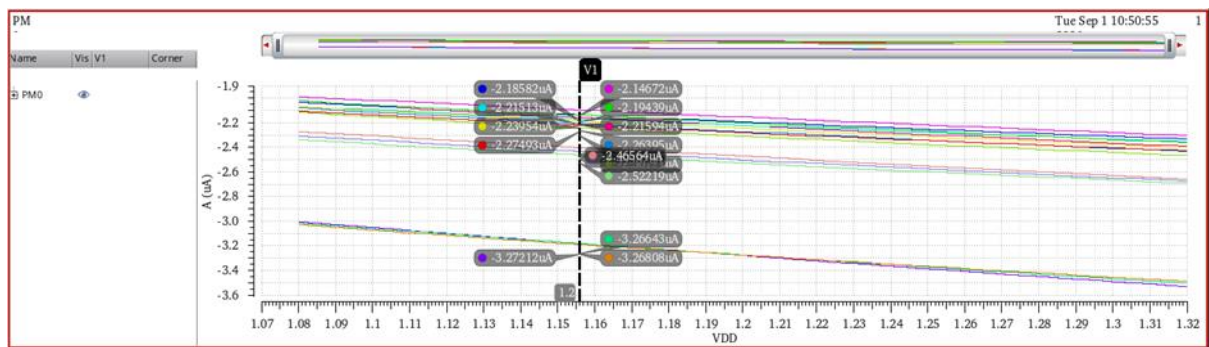
Name	Type	Details	Value	Plot	Save	Spec
id PM0	expr	OP("I/O/PM0" "ids")	-2.73u	☒	☐	
id PM1	expr	OP("I/O/PM1" "ids")	-3.114u	☒	☐	
id PM2	expr	OP("I/O/PM2" "ids")	-2.915u	☒	☐	
id PM3_1	expr	OP("I/O/PM3_1" "ids")	-2.784u	☒	☐	
id NM0	expr	OP("I/O/NM0" "ids")	3.114u	☒	☐	
id NM1_1	expr	OP("I/O/NM1_1" "ids")	2.914u	☒	☐	
id NM2	expr	OP("I/O/NM2" "ids")	2.785u	☒	☐	

Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter v	Filter v	Filter v	Filter v	Filter v	Filter v	Filter v
condition	id PM0				-2.73u	-2.73u
condition	id PM1				-3.432u	-2.905u
condition	id PM2				-3.209u	-2.709u
condition	id PM3_1				-3.168u	-2.521u
condition	id NM0				2.905u	3.432u
condition	id NM1_1				2.708u	3.207u
condition	id NM2				2.522u	3.169u

Line Regulation:



Temp:



Start up:

